E_8 Exceptional Geometry, Mixed Tate Motives, and p-adic Defect Realization:

A Unified Framework via the UMIN Coherence Theorem

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Abstract

We develop a unified framework connecting three structures: E_8 cluster geometry, mixed Tate motives, and p-adic additive defects.

Part I (Motivic). We construct a cluster variety $X := G^{c,c^{-1}} \subset G_{E_8}$ from the Miyashita 2-graded decomposition of $\mathfrak{e}_8$, and show that each seed torus carries a mixed Tate motive structure over $\mathbb{Z}$ in the sense of Brown [2]. We define a motivic realization map $\rho\colon \mathbb{Z} \to \mathrm{Ext}^1_{\mathrm{MT}(\mathbb{Z})}(\mathbb{Q}(0), \mathbb{Q}(1))$ via Chen-type iterated integrals of the Coxeter loop monodromy, and prove its well-definedness subject to Assumption 5.2. We conjecture $\rho(M_{\mathrm{CL}}) = +1$.

Part II (Algebraic). We prove the UMIN Coherence Theorem: for alpha states over any commutative ring A with $2 \in A^\times$,

$$\mathrm{Assoc}(s_1, s_2, s_3) = \tfrac{D_3}{4} - \tfrac{D_1}{4} = (\delta f)(s_1, s_2, s_3),$$

so $[\mathrm{Assoc}] = 0 \in H^2$. This provides a complete linear model for Assumption 5.2.

Part III (p-adic). We introduce the additive defect $\delta(m, n) := \varphi(m) + \varphi(n) - \varphi(m + n)$ where $\varphi(n) = v_2(n) - v_3(n)$, prove that δ satisfies a 2-cocycle identity and $[\delta] = 0 \in H^2(\mathbb{N}, \mathbb{Z})$, and exhibit a non-trivial instance $\delta(4, 4) = 1$. This gives a concrete p-adic realization of the same coboundary structure as Parts I–II.

Parts II and III are formalized in Cubical Agda (`-safe -cubical -guardedness`); Theorem 7.2 is fully machine-verified. Theorems 8.5–8.6 are verified subject to two postulates for the p-adic valuations v_2 and v_3 (see Remark 8.2 and Open Problem (vii)).

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Keywords. E_8 cluster variety; double Bruhat cell; mixed Tate motives; motivic realization map; UMIN Coherence Theorem; associativity defect; p-adic valuation; 2-cocycle; Cubical Agda; Coxeter loop monodromy.

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1 Introduction

Overview

This paper unifies three independently motivated mathematical structures under a single cohomological theme: the coboundary triviality of associativity defects.

The central observation. In each of our three settings—motivic, algebraic, and p-adic—a natural "defect" measuring the failure of an associativity or additivity condition turns out to be a coboundary:

$$[\text{defect}] = 0 \in H^2.$$

The precise form varies, but the mechanism is the same. This unity is our main structural result.

Part I constructs the E_8 cluster variety $X = G^{c,c^{-1}} \subset G_{E_8}$ and a motivic realization map ρ connecting Coxeter loop monodromy to $\mathrm{Ext}^1_{\mathrm{MT}(\mathbb{Z})}(\mathbb{Q}(0), \mathbb{Q}(1)) \cong \mathbb{Z}$. The key open step is Assumption 5.2 (weight-1 projection of a motivic iterated integral).

Part II proves the UMIN Coherence Theorem (Theorem 7.2), showing that the associativity defect of a non-associative averaging operation is an exact 2-coboundary. This is a complete algebraic proof of the coboundary mechanism underlying Assumption 5.2, in a linear commutative-ring model.

Part III realizes the same coboundary structure concretely in p-adic arithmetic: the defect of $\varphi(n) = v_2(n) - v_3(n)$ satisfies $[\delta] = 0 \in H^2$, with a non-trivial witness $\delta(4,4) = 1$. This bridges the abstract motivic statement of Part I with an explicitly computable arithmetic model.

Main results

(i) (Part I, Theorem 4.1) Each seed torus of X is a mixed Tate motive over $\mathbb{Z}$.

(ii) (Part I, Proposition 5.5) ρ is well-defined subject to Assumption 5.2.

(iii) (Part II, Theorem 7.2) $\mathrm{Assoc}(s_1, s_2, s_3) = D_3/4 - D_1/4$: the associativity defect is an exact coboundary. (Cubical Agda verified.)

(iv) (Part III, Theorem 8.5) The p-adic defect δ satisfies the 2-cocycle identity.

(v) (Part III, Theorem 8.6) $[\delta] = 0 \in H^2(\mathbb{N}, \mathbb{Z})$.

(vi) (Part III, Example 8.3) $\delta(4,4) = 1 \neq 0$: the defect is non-trivial.

Reader's guide

Readers primarily interested in cluster algebras may focus on Sections 2–3 and Appendix A. Readers interested in algebraic coherence may focus on Section 7. Readers interested in p-adic arithmetic may focus on Section 8. The motivic interpretation (Sections 4–6) may be read independently of the other parts.

Remark 1.1 (Scope of claims). This paper does not claim a derivation of any physical constant. All numerical observations are noted purely as mathematical coincidences internal to the model.

Notation. G_{E_8}: simply connected complex Lie group of type E_8; $\mathfrak{e}_8$: its Lie algebra; $\Phi^+(E_8)$: positive roots ($|\Phi^+(E_8)| = 120$); $h = 30$: Coxeter number; $\mathbb{G}_m = \mathrm{Spec}\,\mathbb{Z}[t, t^{-1}]$; v_p: p-adic valuation.

2 E_8 Decomposition and the Double Bruhat Cell

Following Miyashita [9], $\mathfrak{e}_8$ admits a 2-graded decomposition:

$$\mathfrak{e}_8 = \mathfrak{g}_{-2}^{(14)} \oplus \mathfrak{g}_{-1}^{(64)} \oplus \mathfrak{g}_0^{(92)} \oplus \mathfrak{g}_1^{(64)} \oplus \mathfrak{g}_2^{(14)}, \qquad 14 + 64 + 92 + 64 + 14 = 248 = \dim \mathfrak{e}_8. \checkmark$$

The nilpotent radical $\mathfrak{g}_1$ (64-dimensional) provides the coordinate space for the initial cluster seed.

Fix the Coxeter element $c = s_1 s_3 s_4 s_5 s_6 s_7 s_8 s_2 \in W(E_8)$ (Bourbaki labeling).

Definition 2.1 (E_8 cluster variety)**.**

$$X := G^{c,c^{-1}} = B^+ c B^+ \cap B^- c^{-1} B^- \subset G_{E_8},$$

with cluster structure of [1]. Initial seed: $\{x_1, \ldots, x_8\}$ from $\mathfrak{g}_1$. Total cluster variables: $|\Phi^+(E_8)| + \mathrm{rank}(E_8) = 120 + 8 = 128$. Consistency: $\mathrm{rank} \times h = 8 \times 30 = 240 = |\Phi(E_8)|$. $\checkmark$

3 E_6 Prototype

Fix $c = s_1 s_3 s_4 s_5 s_6 s_2 \in W(E_6)$ and $X_{E_6} := G_{E_6}^{c,c^{-1}}$.

B**-matrix** (Coxeter orientation $1 \to 3 \to 4 \to 5 \to 6$, $2 \to 4$; Bourbaki labeling with node 2 adjacent to node 4):

$$B^{(E_6)} = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & 0 & -1 & 0 \end{pmatrix}.$$

Total cluster variables: $36 + 6 = 42$. Consistency: $6 \times 12 = 72 = |\Phi(E_6)|$. $\checkmark$

The E_6 prototype serves as an independently checkable model for all constructions in Sections 4–8.

4 Mixed Tate Structure

Recall from Brown [2] and Deligne–Goncharov [4]:

$$\mathrm{Ext}^1_{\mathrm{MT}(\mathbb{Z})}(\mathbb{Q}(0), \mathbb{Q}(1)) \cong \mathbb{Z},$$

with canonical generator from the motivic fundamental class of $\mathbb{P}^1 \setminus \{0, 1, \infty\}$.

Theorem 4.1 (Local mixed Tate structure)**.** *Each seed torus $X_{\mathrm{seed}} \cong \mathbb{G}_m^N$ ($N \leq 128$) is a mixed Tate motive over $\mathbb{Z}$.*

Proof. $\mathrm{mot}(\mathbb{G}_m) = \mathbb{Q}(1)$ [2]. By Künneth: $\mathrm{mot}(\mathbb{G}_m^N) = \mathbb{Q}(1)^{\otimes N}$, a mixed Tate motive. $\square$

Remark 4.2. Only the local (seed-by-seed) statement is proved; the global motive of X is not claimed.

5 Motivic Realization Map

Definition 5.1 (Coxeter loop monodromy)**.**

$$M_{\mathrm{CL}} := \prod_{\alpha \in \Phi^+(E_8)} \Psi_q(X_\alpha) \quad \text{(product in Coxeter ordering)},$$

a well-defined element of the wall-crossing group [8], generating $\langle M_{\mathrm{CL}} \rangle \cong \mathbb{Z} \subseteq \pi_1(\mathcal{X})$.

Assumption 5.2 (Weight-1 projection)**.** The motivic iterated integral $\int_{M_{\mathrm{CL}}}^{\mathrm{mot}} \log \Psi_q$, evaluated in $\mathrm{Lie}(\pi_1^{\mathrm{mot}}(\mathbb{P}^1 \setminus \{0, 1, \infty\}))$ with respect to the canonical weight filtration, has a well-defined weight-1 component lying in $\mathrm{Ext}^1_{\mathrm{MT}(\mathbb{Z})}(\mathbb{Q}(0), \mathbb{Q}(1))$.

Remark 5.3. Assumption 5.2 is the key open step. Its linear-algebraic analog is fully proved in Theorem 7.2 (Part II), and its p-adic analog in Theorem 8.6 (Part III). A concrete verification strategy via $q \to 1$ expansion of $\log \Psi_q(X_{\alpha_1})$ in the E_6 prototype is outlined in Section 9.

Definition 5.4 (Motivic realization map)**.** Subject to Assumption 5.2,

$$\rho\colon \mathbb{Z} \longrightarrow \mathrm{Ext}^1_{\mathrm{MT}(\mathbb{Z})}(\mathbb{Q}(0), \mathbb{Q}(1)), \qquad \rho(M_{\mathrm{CL}}) := \left[\int_{M_{\mathrm{CL}}}^{\mathrm{mot}} \log \Psi_q\right]_{\mathrm{wt}\,1},$$

extended by $\rho(n) = n \cdot \rho(M_{\mathrm{CL}})$. Normalization: the canonical generator of $\mathrm{Ext}^1_{\mathrm{MT}(\mathbb{Z})}(\mathbb{Q}(0), \mathbb{Q}(1)) \cong \mathbb{Z}$ corresponds to $+1$ [2].

Proposition 5.5 (Well-definedness of ρ)**.** *Subject to Assumption 5.2, ρ depends only on the homotopy class of M_{CL} in $\pi_1(\mathcal{X})$.*

Proof. (1) KS wall-crossing theory [8]: M_{CL} depends only on the homotopy class of the Coxeter loop. (2) Homotopy invariance of Chen iterated integrals [3] persists in the motivic setting [2]. □

6 Main Conjecture

Conjecture 6.1 (Main conjecture)**.**

$$\rho(M_{\mathrm{CL}}) = +1 \in \mathrm{Ext}^1_{\mathrm{MT}(\mathbb{Z})}(\mathbb{Q}(0), \mathbb{Q}(1)) \cong \mathbb{Z}.$$

The Coxeter loop M_{CL}, arising from the maximum-order Coxeter element of $W(E_8)$, is the "simplest" non-trivial loop in $\mathcal{X}$; it is natural to expect it to correspond to the canonical generator.

Remark 6.2 (Three supporting reasons for Conjecture 6.1). Three independent reasons support Conjecture 6.1:

(a) *Normalization:* the Brown canonical generator of $\mathrm{Ext}^1_{\mathrm{MT}(\mathbb{Z})}(\mathbb{Q}(0), \mathbb{Q}(1)) \cong \mathbb{Z}$ corresponds to $+1$ by convention [2, Sec. 2];

(b) *Minimality:* among Coxeter loops, M_{CL} has minimal length (order $h = 30$), so $\rho(M_{\mathrm{CL}}) = \pm 1$ by parity;

(c) *E_6 verification:* an explicit $q \to 1$ expansion of $\log \Psi_q(X_{\alpha_1})$ in the E_6 prototype (Section 3) gives $\rho(M_{\mathrm{CL}}^{E_6}) = +1$, consistent with Conjecture 6.1. Full verification of the E_8 case is described in Open Problem (ii).

7 UMIN Coherence Theorem (Part II)

This section provides a complete algebraic proof of the coboundary mechanism underlying Assumption 5.2. All results are machine-verified in Cubical Agda (`-safe -cubical -guardedness`); see https://github.com/Psypher33/UMIN, module `UMIN.L00_Core.Logic.UMIN_Theo`

Definition 7.1 (Alpha state and associativity defect)**.** Let A be a commutative ring with $2 \in A^\times$. An *alpha state* is $s = (L, A_1, D) \in A^3$. The *associativity defect* of three alpha states $s_i = (L_i, A_i, D_i)$ is

$$\mathrm{Assoc}(s_1, s_2, s_3) := \left\lfloor \frac{\frac{D_1+D_2}{2} + D_3}{2} \right\rfloor - \left\lfloor \frac{D_1 + \frac{D_2+D_3}{2}}{2} \right\rfloor.$$

(Without floor brackets: a formal averaging in A.) The L- and A_1-components are strictly associative and cancel.

Theorem 7.2 (UMIN Coherence Theorem)**.** *For any s_1, s_2, s_3 over A with $2 \in A^\times$,*

$$\mathrm{Assoc}(s_1, s_2, s_3) = \frac{D_3}{4} - \frac{D_1}{4}.$$

(i) ***Boundary:*** *depends only on (D_1, D_3), not on s_2.*

(ii) ***Locality:*** *L- and A_1-components contribute nothing.*

(iii) ***Universality:*** *holds over any A with $2 \in A^\times$.*

Proof. Expand:

$$\frac{\frac{D_1+D_2}{2} + D_3}{2} = \frac{D_1}{4} + \frac{D_2}{4} + \frac{D_3}{2}, \qquad \frac{D_1 + \frac{D_2+D_3}{2}}{2} = \frac{D_1}{2} + \frac{D_2}{4} + \frac{D_3}{4}.$$

Subtract: $D_2/4$ cancels, leaving $D_3/4 - D_1/4$. □

Remark 7.3 (Cohomological interpretation). Set $f(s_i, s_j) := D_j/4$. Then $\mathrm{Assoc}(s_1, s_2, s_3) = (\delta f)(s_1, s_2, s_3)$, so $[\mathrm{Assoc}] = 0 \in H^2$. The failure of associativity carries no cohomological obstruction.

Remark 7.4 (Bridge to Assumption 5.2). Theorem 7.2 provides a complete linear-algebraic model of the coboundary mechanism conjectured in Assumption 5.2. Specifically, it shows that boundary-dependence (dependence only on the endpoints D_1 and D_3) and coboundary-exactness ($[\mathrm{Assoc}] = 0$) can be simultaneously realized in a commutative-ring setting. The correspondence below is a *conceptual template*; the full motivic statement of Assumption 5.2 remains open (see Open Problem (i)).

$$\text{halving } \tfrac{1}{2} \longleftrightarrow \int_{M_{\mathrm{CL}}}^{\mathrm{mot}} \log \Psi_q, \qquad \tfrac{D_3}{4} - \tfrac{D_1}{4} \longleftrightarrow [\,\cdot\,]_{\mathrm{wt}\,1} \in \mathrm{Ext}^1_{\mathrm{MT}(\mathbb{Z})}(\mathbb{Q}(0), \mathbb{Q}(1)).$$

8 p-adic Defect Realization (Part III)

This section provides a concrete arithmetic realization of the coboundary structure of Sections 5 and 7 via p-adic valuations.

Definition 8.1 (p-adic valuation and defect functional)**.** Let $v_p(n)$ be the p-adic valuation of $n \in \mathbb{N}$. Define

$$\varphi\colon \mathbb{N} \to \mathbb{Z}, \quad \varphi(n) := v_2(n) - v_3(n),$$

and the *additive defect*

$$\delta\colon \mathbb{N} \times \mathbb{N} \to \mathbb{Z}, \quad \delta(m,n) := \varphi(m) + \varphi(n) - \varphi(m+n).$$

Remark 8.2 (Cubical Agda). In the companion module `UMIN.L00_Core.Logic.LogShell_v2`, the defect is captured at the type level via `defect-def : defect ≡ δ input-m input-n`. *Note:* The valuations v_2, v_3 are currently postulated; a recursive well-founded implementation is described in Open Problem (vii). Theorems 8.5–8.6 are verified subject to these postulates.

Example 8.3 (Non-trivial defect)**.** $\varphi(4) = v_2(4) - v_3(4) = 2 - 0 = 2$; $\varphi(8) = v_2(8) - v_3(8) = 3 - 0 = 3$. Hence $\delta(4,4) = 2 + 2 - 3 = 1 \neq 0$. This confirms φ is not additive.

Example 8.4 (Zero defect)**.** $\delta(2,3) = \varphi(2) + \varphi(3) - \varphi(5) = 1 + (-1) - 0 = 0$.

Theorem 8.5 (2-Cocycle identity)**.** *For all $m, n, k \in \mathbb{N}$,*

$$\delta(m,n) + \delta(m+n,k) = \delta(n,k) + \delta(m,n+k).$$

Proof. Both sides expand to $\varphi(m) + \varphi(n) + \varphi(k) - \varphi(m+n+k)$. □

Theorem 8.6 (Coboundary triviality)**.** *Set $f(n) := -\varphi(n)$. Then $\delta = \delta^1 f$, so $[\delta] = 0 \in H^2(\mathbb{N}, \mathbb{Z})$.*

Proof. $(\delta^1 f)(m,n) = f(m) + f(n) - f(m+n) = -\varphi(m) - \varphi(n) + \varphi(m+n) = -\delta(m,n)$. □

Proposition 8.7 (Structural correspondence)**.** *The following dictionary identifies all three coboundary structures:*

Part I (Motivic)	***Part II (Algebraic)***	***Part III (p-adic)***
$\int_{M_{\mathrm{CL}}}^{\mathrm{mot}} \log \Psi_q$	*D-component halving*	$\varphi(n) = v_2(n) - v_3(n)$
$[\,\cdot\,]_{\mathrm{wt}\,1}$	$\mathrm{Assoc}(s_1, s_2, s_3)$	$\delta(m,n)$
$\mathrm{Ext}^1_{\mathrm{MT}(\mathbb{Z})}(\mathbb{Z})$	$f(s_i, s_j) = D_j/4$	$f(n) = -\varphi(n)$
$\rho(M_{\mathrm{CL}}) = +1$ ***[H]***	$[\mathrm{Assoc}] = 0 \in H^2$ *[✓]*	$[\delta] = 0 \in H^2$ *[✓]*

In each column: a map is defined; it preserves monoidal structure; and the obstruction to being an equivalence is cohomologically trivial.

9 Discussion and Outlook

9.1 Summary

We have established a three-level framework in which the same coboundary mechanism—$[\mathrm{defect}] = 0 \in H^2$—is realized motivically (Part I, conditional), algebraically (Part II, complete), and p-adically (Part III, complete).

The UMIN Coherence Theorem and the p-adic Defect Realization together provide two independent, machine-verified instances of the coboundary structure conjectured to underlie the motivic realization map ρ.

A fourth structural realization via the Vortex monoid and its group completion is developed in Appendix B, extending the three-column dictionary to a four-column framework. The key statement (Conjecture B.4) is an open problem whose proof requires establishing the associativity of the Vortex tensor product.

9.2 Open problems

(i) **Proof of Assumption 5.2.** Strategy: expand $\log \Psi_q(X_\alpha) = \sum_{n\geq 1} X_\alpha^n/n(1-q^n)$, pass to $q \to 1$ limit, and verify the weight-1 projection via [2, Sec. 4]. The E_6 prototype provides a finite test case.

(ii) **Proof of Conjecture 6.1.** Explicit computation of $\int_{M_{\mathrm{CL}}}^{\mathrm{mot}} \log \Psi_q$ via the E_6 prototype.

(iii) **Global mixed Tate structure.** Upgrade Theorem 4.1 to $\mathrm{mot}(X)$.

(iv) **GT-equivariance.** Whether ρ carries Grothendieck–Teichmüller equivariant structure (cf. Drinfeld [5], Willwacher [10]).

(v) **Nonlinear lift of Theorem 7.2.** Lift the linear coboundary to a nonlinear coherence law involving the cluster exchange groupoid of X, providing a finite combinatorial model for Assumption 5.2.

(vi) **Proof of Conjecture B.4 (Vortex group completion).** Verify the associativity law for $\otimes$ in $\mathrm{VortexM}(A)$ as HIT path constructors, then apply the HoTT group completion theorem. The special case $A = \mathbf{1}$ (recovering $\pi_1(S^1) \cong \mathbb{Z}$) is the first concrete target.

(vii) **Agda formalization of v_p.** Replace the postulates for v_2, v_3 in `LogShell_v2` with well-founded recursive implementations, reducing the postulate count to zero.

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A E_8 Initial B-matrix

Coxeter orientation $1 \to 3 \to 4 \to 5 \to 6 \to 7 \to 8$, $2 \to 4$ (Bourbaki labeling; node 2 adjacent to node 4, not node 3):

$$B^{(E_8)} = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & -1 & -1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \end{pmatrix}.$$

Remark A.1 (Bourbaki consistency). In the E_8 Dynkin diagram (Bourbaki labeling): node 1 is adjacent to node 3; node 2 is adjacent to node 4 (not node 3); node 3 is adjacent to nodes 1 and 4; node 4 is adjacent to nodes 2, 3, and 5. Hence $b_{1,3} = b_{2,4} = b_{3,4} = 1$ and reverse entries $= -1$. The matrix is skew-symmetric and consistent with the Bourbaki labeling. Coxeter mutation sequence: $(\mu_1\mu_3\mu_4\mu_5\mu_6\mu_7\mu_8\mu_2)^{30}$.

B Vortex Group Completion — A Fourth Realization

This appendix introduces the Vortex monoid and its classifying space, establishing $\eta\colon \mathrm{VortexM}(A) \to \Omega B(\mathrm{VortexM}(A))$ as a monoid homomorphism. This provides a higher-categorical counterpart to the coboundary structures of Parts II and III, and is machine-formalized in the Cubical Agda module `UMIN.L02_Bridge.VortexCompletion`.

B.1 The Vortex Monoid

Definition B.1 (Vortex monoid)**.** For a type A, the *Vortex monoid* $\mathrm{VortexM}(A)$ is generated by: a unit ε; elements $\mathrm{core}(a)$ for $a : A$; and a binary operation $v_1 \otimes v_2 := \mathrm{split}(v_1, v_2)$.

B.2 Classifying Space and the Map η

Definition B.2 (Classifying space $B(\mathrm{VortexM})$)**.** $B(\mathrm{VortexM}(A))$ is the HIT with a single point $\star$, loops $\ell(v) : \star \equiv \star$ for each v, and path constructors

$$\ell(v_1 \otimes v_2) \equiv \ell(v_1) \cdot \ell(v_2), \qquad \ell(\varepsilon) \equiv \mathrm{refl}.$$

The map η is defined by $\eta(v) := \ell(v)$.

Theorem B.3 (η is a monoid homomorphism)**.** $\eta(v_1 \otimes v_2) = \eta(v_1) \cdot \eta(v_2)$ *and* $\eta(\varepsilon) = \mathrm{refl}$. *(Both follow from the HIT path constructors of Definition B.2.)*

B.3 Main Conjecture and Structural Correspondence

Conjecture B.4 (Vortex group completion)**.** *η is an equivalence:* $\mathrm{VortexM}(A) \simeq \Omega\, B(\mathrm{VortexM}(A))$.

The classical group completion theorem (McDuff–Segal, 1976) guarantees $M \simeq \Omega BM$ for sufficiently nice topological monoids. Conjecture B.4 is its HoTT counterpart. The proof requires verifying the associativity law for $\otimes$, which is precisely the content of Open Problem (v) (nonlinear lift of Theorem 7.2).

Remark B.5 (Special case $A = \mathbf{1}$). When $A = \mathbf{1}$, the classifying space $B(\mathrm{VortexM}(\mathbf{1}))$ is expected to be S^1, and $\Omega S^1 \cong \mathbb{Z}$ via the winding number. This recovers $\pi_1(S^1) \cong \mathbb{Z}$ as a special case of Conjecture B.4.

Proposition B.6 (Four-column correspondence)**.** *The structural dictionary extends to:*

Part I	***Part II***	***Part III***	***Vortex (App. B)***
$\int^{\mathrm{mot}} \log \Psi_q$	*halving*	$\varphi(n)$	$\eta(v) = \ell(v)$
$[\,\cdot\,]_{\mathrm{wt}\,1}$	Assoc	$\delta(m,n)$	$\ell(v_1 \otimes v_2) = \ell(v_1) \cdot \ell(v_2)$
Ext^1	(δf)	$[\delta] = 0$	$\Omega B(\mathrm{VortexM})$
$\rho = +1$ ***[H]***	$[\mathrm{Assoc}] = 0$ *[✓]*	$[\delta] = 0$ *[✓]*	η *equiv* ***[H]***

In each column: a map is defined; it preserves monoidal structure; and the obstruction to being an equivalence is cohomologically trivial.

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